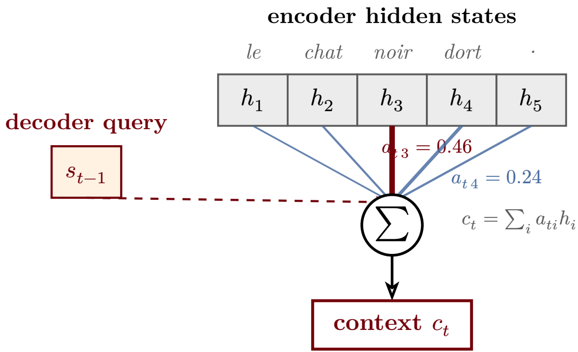
Bahdanau / Luong cross-attention

decoder query scores every encoder state · softmax → alignment weights · weighted-sum context vector



align weights: $a_{ti} = \mathrm{softmax}_i(e_{ti})$

Bahdanau (additive): $e_{ti} = v^\top \tanh(W_s s_{t-1} + W_h h_i)$

Luong (mult.): $e_{ti} = s_{t-1}^\top h_i$ or $s_{t-1}^\top W h_i$

alignment A_{ti}

target steps t (English)	*le*	*chat*	*noir*	*dort*	*.*
the	**0.62**	0.18	0.08	0.06	0.06
black	0.10	0.20	**0.58**	0.07	0.05
cat	0.07	0.11	**0.46**	0.24	0.12
sleeps	0.05	0.07	0.12	**0.70**	0.06
<eos>	0.06	0.06	0.06	0.14	**0.68**

source positions i (French)

The "cat" row of the heatmap (highlighted in panel A) shows the alignment weights for one decoder step; edge thickness ∝ weight.